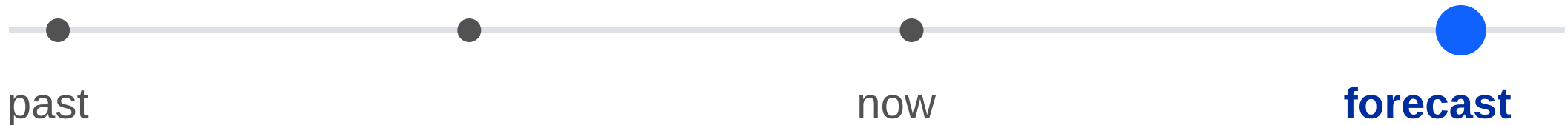


CE436 / PROJECT PROPOSAL

# TEMPERATURE FORECAST ON FPGA

**+3 h** from past hourly values



3 students • 12 weeks • C++ + Verilog/SystemVerilog

# A testable forecasting task

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## INPUT

Hourly temperature  
at one location



## OUTPUT

One temperature  
3 hours ahead

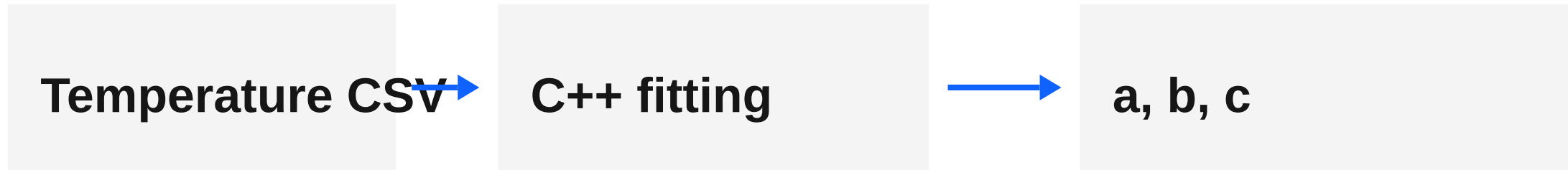
Known through  $t$

**Forecast for  $t + 3\text{ h}$**

# C++ trains. FPGA forecasts.

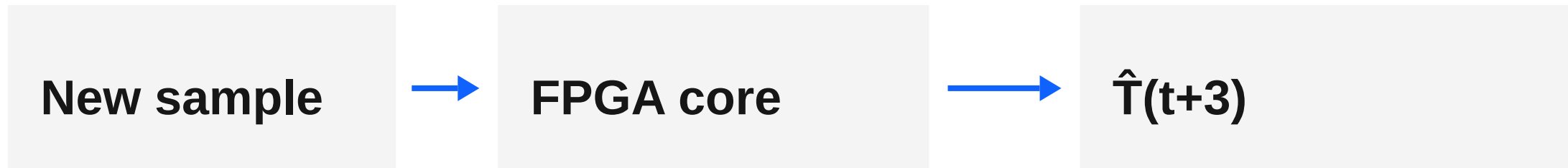
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## OFFLINE ON PC



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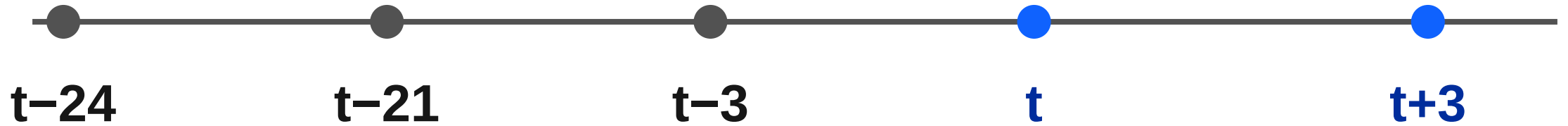
## RUNTIME ON FPGA



The coefficients stay fixed while the FPGA runs.

# Four past values. One future target.

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$$t + 3 - 24 = t - 21$$

25 consecutive samples are stored; the future target is never an input.

# A linear model from past values

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$$\begin{aligned}\hat{T}(t+3) = & T(t-21) \\ & + a[T(t)-T(t-24)] \\ & + b[T(t)-T(t-3)] & + c\end{aligned}$$

---

## ANCHOR

same hour yesterday

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## DAY-TO-DAY

current vs. yesterday

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## RECENT CHANGE

latest 3 hours

# How is one forecast computed?

ILLUSTRATIVE VALUES — NOT A MEASURED RESULT

|             |             |             |             |
|-------------|-------------|-------------|-------------|
| <b>19°C</b> | <b>21°C</b> | <b>18°C</b> | <b>20°C</b> |
| t-24        | t-21        | t-3         | t           |

a = 0.5 b = 0.2

|             |             |             |                                                                                      |               |
|-------------|-------------|-------------|--------------------------------------------------------------------------------------|---------------|
| <b>21.0</b> | <b>+0.5</b> | <b>+0.4</b> |  | <b>21.9°C</b> |
| anchor      | day shift   | 3 h trend   |                                                                                      |               |

subtract → multiply → add

# Train on the past. Test on the future.

---

$$x1 = T(t) - T(t-24)$$

$$x2 = T(t) - T(t-3)$$

$$y = a \cdot x1 + b \cdot x2 + c$$

fit a, b, c by least squares

## 01 TRAIN

fit coefficients

## 02 VALIDATE

select settings

## 03 TEST

evaluate once

Split chronologically; training labels must not cross into later periods.

# Does the model beat simple forecasts?

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CURRENT VALUE

$T(t)$

SAME HOUR YESTERDAY

$T(t-21)$

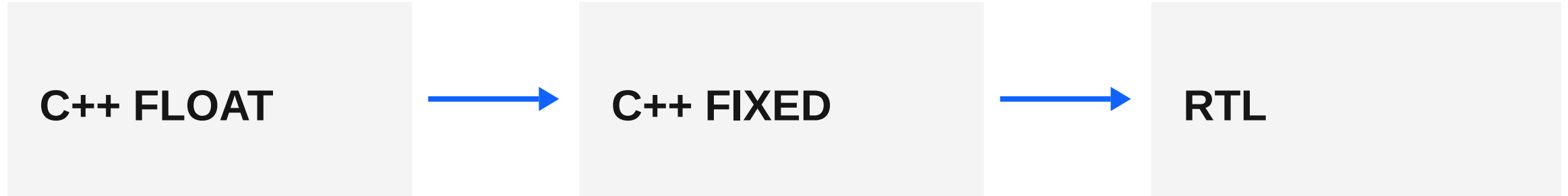
LINEAR MODEL

$\hat{T}(t+3)$

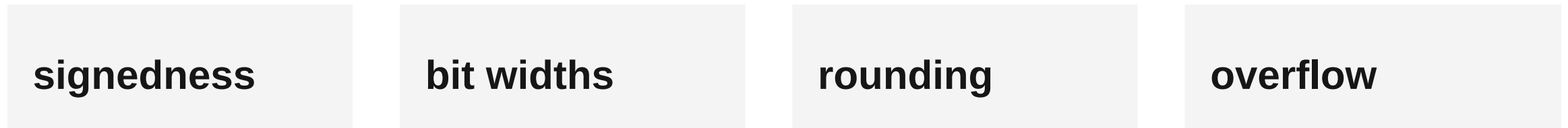
Compare MAE on the same held-out test timestamps.

# Fixed-point needs an explicit contract

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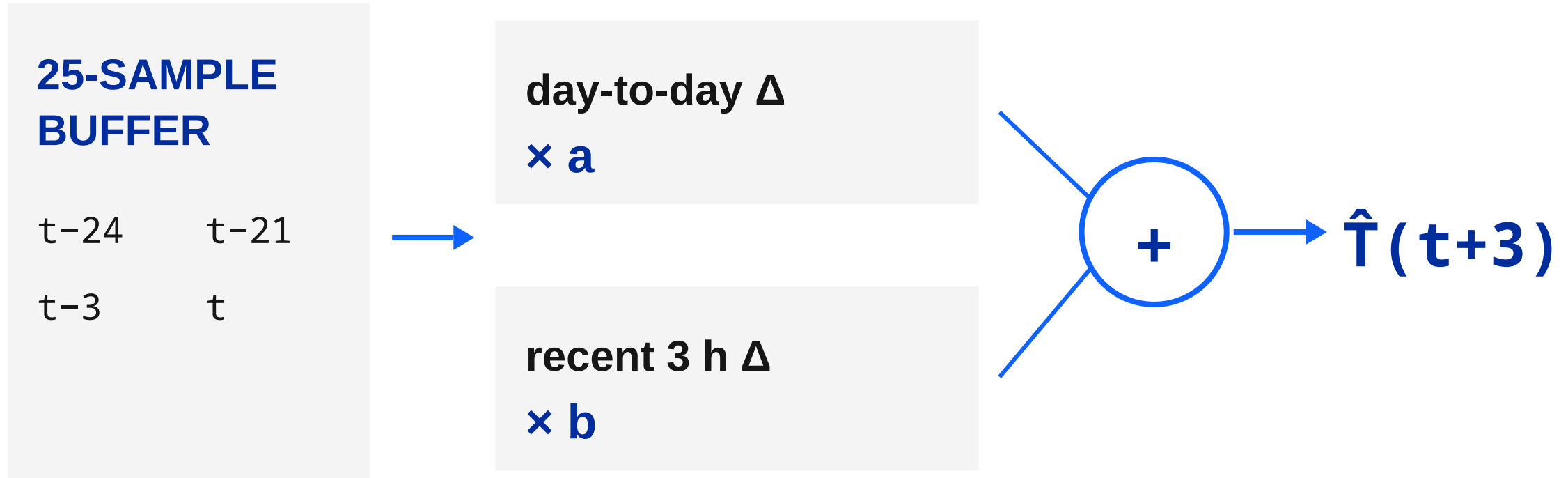


**Define the arithmetic before coding RTL:**



Forecast error and quantization error are different measurements.

# From sample history to arithmetic



Accept a sample on valid; output only after the buffer is ready.

# Three verification layers

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**01**

**C++**

float vs. fixed

---

quantization

**02**

**TESTBENCH**

RTL vs. reference

---

bit-exact output

**03**

**BOARD**

integrated run

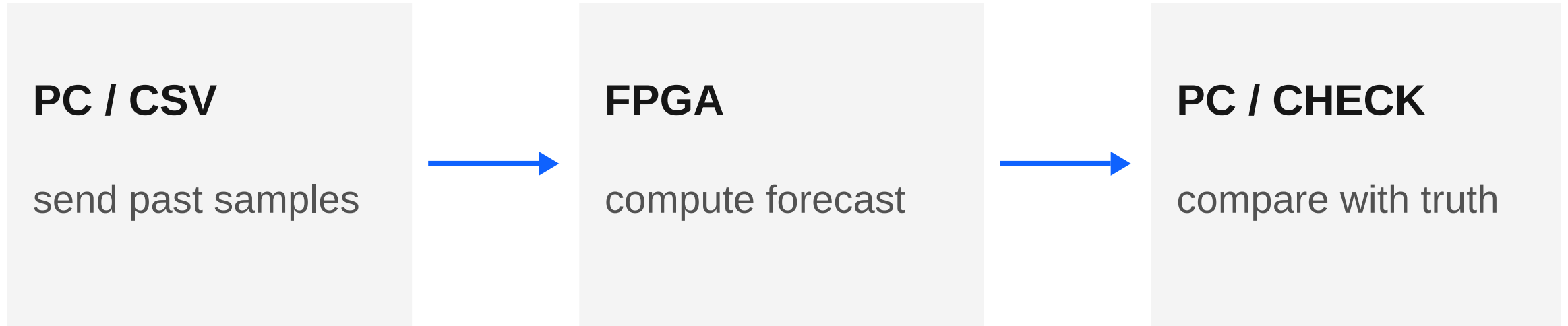
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I/O and sample order

All layers use the same input vectors and arithmetic specification.

# Replay historical data for the demo

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**Only samples through t enter the FPGA.**

# 12 weeks. Prototype early. Evaluate deeply.

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**W1–3**

**PROTOTYPE**

data + C++  
working RTL core

**W4–6**

**VALIDATE**

model vs. baselines  
fixed-point sweep

**W7–9**

**HARDEN**

bit-exact tests  
resource trade-offs

**W10–12**

**DEMO**

board integration  
final test + report

**Each phase ends with a measurable deliverable.**

# Three students. Clear ownership.

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## STUDENT 1

### DATA + C++

dataset + MAE  
coefficients

---

missing hours

## STUDENT 2

### FIXED + RTL

RTL core  
synthesis

---

board TBD

## STUDENT 3

### VERIFY + DEMO

test vectors  
interface + logs

---

high error

**One shared contract: data, coefficients, rounding, valid, and latency.**

# One forecast. Two things to prove.

---

## 01 FORECAST QUALITY

Beat the baselines  
on unseen data

## 02 HARDWARE CORRECTNESS

Match the C++ fixed-  
point reference

### **Feedback needed:**

scope • FPGA board/tools • minimum demo